RFID Wisp Terminal

User Manual



Read and write the filament tags of the QIDI box at the terminal, see what is in the box and manage spools with Spoolman – stand-alone, no PC needed.

Applies to RFID Wisp Terminal version 0.8.1 (as of October 2026)

Note: This English manual is an automated translation of the German original. Mistakes cannot be ruled out. If something is unclear, please compare it with the German manual.

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1 Introduction

The RFID Wisp Terminal is a stand-alone device for reading and writing the RFID tags of the filament spools of QIDI multi-colour boxes. It needs no PC: it works over Wi-Fi, shows on its touch display what is in the four slots of a QIDI box and works together with Spoolman. The tags are the same as those of the RFIDwisp app; both programs can read and write the same tags.

The terminal consists of a WT32-SC01 Plus (ESP32-S3 with a 3.5-inch touch display) and an RC522 RFID module. The tags are MIFARE Classic 1K tags.

This manual describes the firmware from version 0.8.0 on. It shows the operation step by step, with pictures of the user interface.

1.1 What you need

- A terminal (WT32-SC01 Plus with the RC522 module), built and flashed with the firmware.
- A Wi-Fi network in which the terminal and your printer can be reached.
- For the box overview, a QIDI Plus4 or Max4 (also with the newer firmware: Plus4 2.1.1, Max4 01.01.07.01). The QIDI Q2 is not supported.
- Optional: a Spoolman server and the RFIDwisp app's Klipper module rfid_bridge on the printer (it provides the spool numbers of the slots).

To build the terminal yourself you need the following parts (bill of materials):

- 1 × WT32-SC01 Plus (Wireless-Tag Technology Co., Limited)
- 1 × RC522 RFID module (MFRC-522)
- 1 × KCD11 2-pin 10 × 15 mm mini push button switch
- 1 × USB 3.1 Type-C jack, 4 pin, welding wire, 3 A, female, straight, DIY connector
- 1 × USB 3.1 Type-C male plug for cable end
- 1 × 3 mm LED, blue
- 1 × 680 Ω resistor (R680)
- Wires AWG24 for the connection between WT32 and RC522
- Wires AWG18 or 0.75 mm² for the power lines

The files for you to print the housing can be found here: RFID Wisp CAD

1.2 Conventions in this manual

- **Bold** text is a label from the terminal screen (buttons and fields).
- `Monospaced text` stands for addresses, file names and commands.

2 First start

The firmware is loaded onto the terminal from a computer over USB. The files and the steps are in the file `FLASHING.md` in the release archive at `https://github.com/ThorSc/RFIDwisp-Terminal/releases/latest`. Later versions install themselves (see the section on firmware updates).

2.1 Setting up Wi-Fi

On the first start the terminal does not know your Wi-Fi yet. It opens a Wi-Fi network of its own named RFIDwisp-Setup.

1. Connect a smartphone or laptop to the Wi-Fi network RFIDwisp-Setup. The setup page opens by itself; otherwise open `http://192.168.4.1` in a browser.
2. Choose your Wi-Fi network and enter its password.
3. Enter the address of your server under Spoolman, for example `http://192.168.1.100:7912`. Without Spoolman leave the field empty; you can add it later in the settings.
4. If you like, set a password for updates over the network (OTA).
5. Save. The terminal restarts and connects to your Wi-Fi network.

After that the main screen appears. You can set up the Wi-Fi again later in the settings.

2.2 Adding a printer

For the main screen to show the slots of your QIDI box, add at least one printer in the settings under Printer (section Settings).

3 The main screen

The main screen shows what is in the four slots of the selected QIDI box. It is also the starting point for editing tags.



Figure 1: The main screen with one printer and one box

- At the top are the name of the firmware with its version and the gear button for the settings.
- Below is the printer with its address. The button on the right reads the box again.
- The table shows one slot per row.
- Edit tag opens the editing of a tag.
- The status line at the very bottom reports what the terminal is doing or what did not work.

3.1 The table of slots

Column	Meaning
Dot and number	The state of the slot and its number in the box (0 to 3). Green: filament loaded. Turquoise: this slot is feeding the extruder right now. Red: empty.
Material	The material the box reports for the slot, for example PLA Basic.
Vendor (no heading)	The QIDI vendor of the tag: Gen. stands for Generic, QIDI for QIDI.
Colour field	The colour of the filament.
No.	The Spoolman spool number stored on the tag. It only appears if the Klipper module rfid_bridge runs on the printer.
Vendor (second column)	The name of the vendor in Spoolman. It only appears with Spoolman.
Weight	The remaining weight of the spool in grams according to Spoolman. It only appears with Spoolman.

3.2 Several printers and boxes

If you have added more than one printer, a list for choosing the printer appears at the top. If the printer has more than one box, a second list for the box is added. After switching, the terminal reads the slots again straight away. The chosen printer is kept in the device.



Figure 2: Choosing printer and box

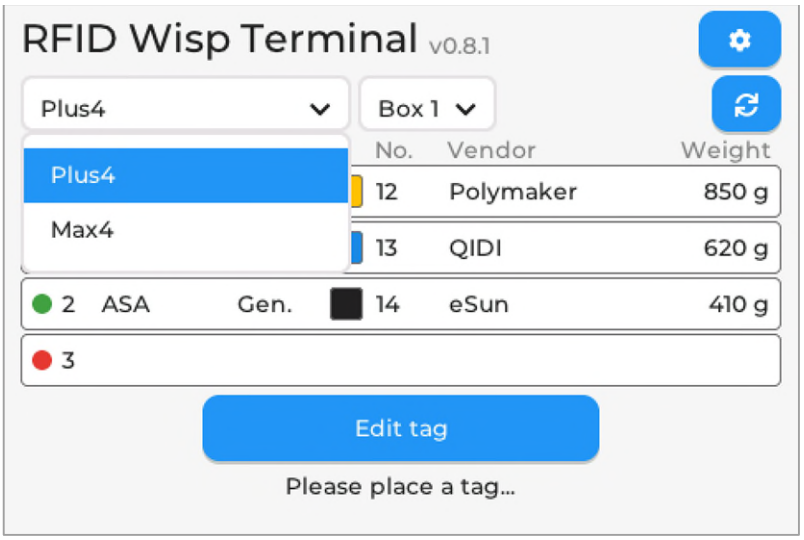


Figure 3: The opened list of printers

3.3 When the printer does not answer

If the printer cannot be reached, not connected is shown in red in place of the table, with the reason below it. Check that the printer is switched on and that the address in the settings is correct. The refresh button tries again.

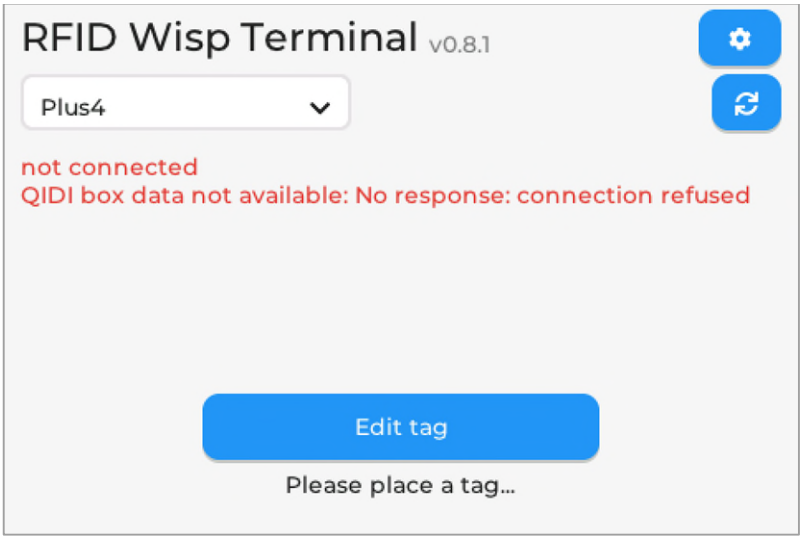


Figure 4: The printer cannot be reached

If the printer answers but has no QIDI box data (for example a QIDI Q2 or a printer of another make), not supported is shown in place of the table, with an explanation below it. The connection itself is fine in that case, so do not look for the fault in the network.

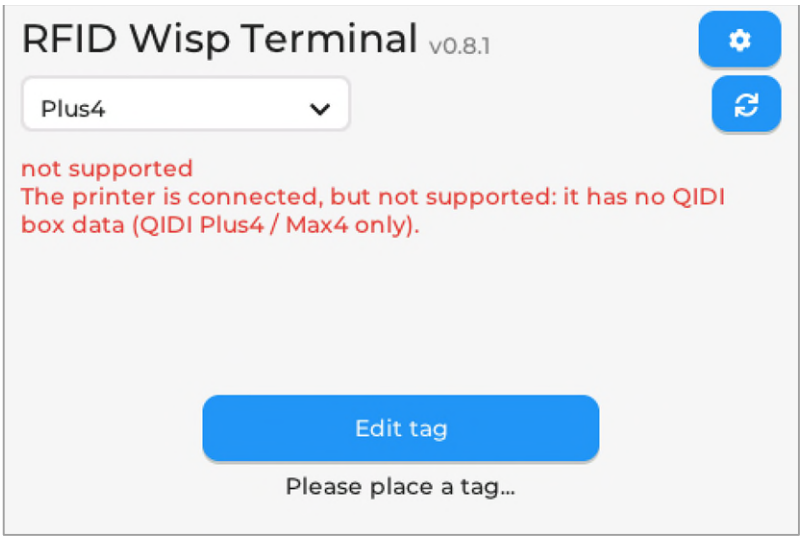


Figure 5: The printer is not supported

4 Reading and writing tags

4.1 Reading a tag

Put the tag on the RFID module while the main screen is showing. The terminal recognises it by itself, reads it and shows its data in the editing screen. You do not have to press a button. Afterwards take the tag off the module before the same tag is to be read again.

If the tag is blank or holds no valid spool data, the editing screen opens and the status line gives the reason. The fields then show the values used last (default values after switching on). Set the data you want and write the tag.

4.2 The editing screen

The editing screen shows the data of the tag. Edit tag on the main screen opens it even if no tag is lying on the reader. The fields then show the values used last, default values after switching on.

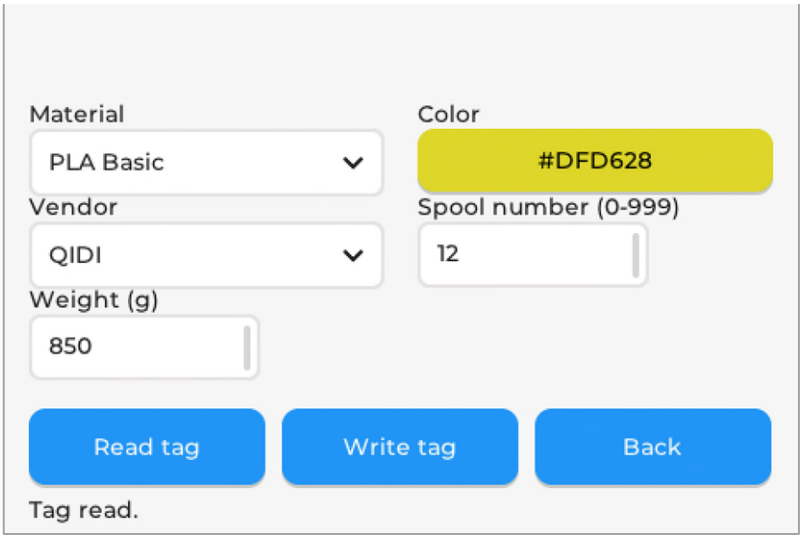


Figure 6: The editing screen without Spoolman

Field	Meaning
Material	The filament material. Tap the field and choose from the list.
Colour	The colour. Tap the field to choose it (see below).
Vendor	The QIDI vendor, GENERIC or QIDI.
Spool number (0-999)	The number of the spool. With Spoolman it is the Spoolman spool number.
Weight (g)	The weight in grams, 0 to 10000.

Three buttons are at the bottom. Read tag reads another tag without going back to the main screen. Write tag writes the data shown onto a tag. Back returns to the main screen.

4.3 Choosing the colour

The colour field opens an overview of the colours that QIDI boxes know. Tap a colour to take it.



Figure 7: The colour selection

4.4 Entering numbers

Tapping the field for the spool number or the weight opens a number pad. Enter the number and confirm with the check mark. Values that are too large are cut to the highest allowed value.

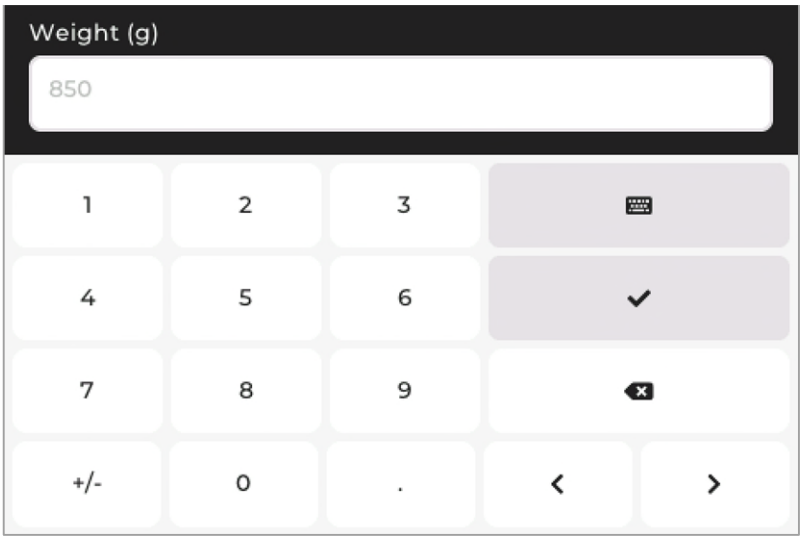


Figure 8: The number pad

4.5 Writing a tag

1. Set the data in the editing screen, or read an existing tag first.
2. Tap Write tag and hold the target tag on the RFID module.
3. Wait for the message Tag written. After writing, the terminal reads the tag once more and compares it with the data.

A tag that has just been read and is still on the module can be written directly; you do not have to take it off first.

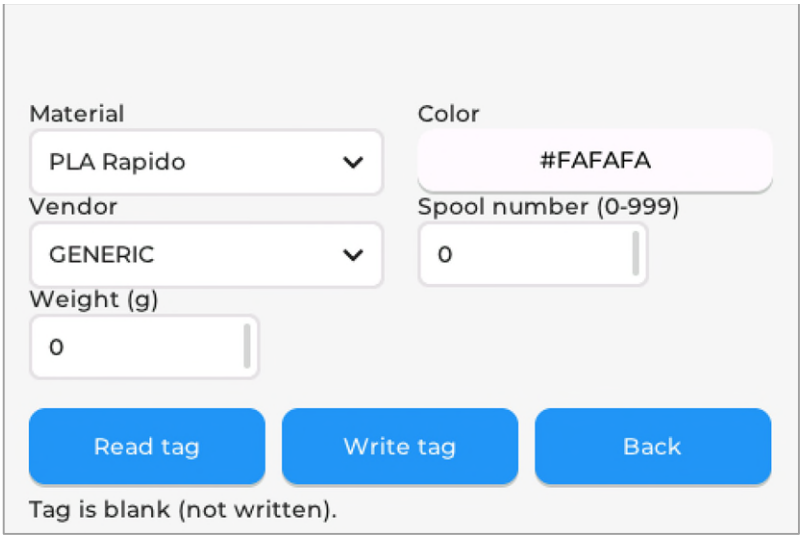


Figure 9: A blank tag: the editing screen with the default values and the message

5 Working with Spoolman

If Spoolman is set up and switched on, the editing screen changes: instead of the fields for vendor and spool number you choose a spool from Spoolman.

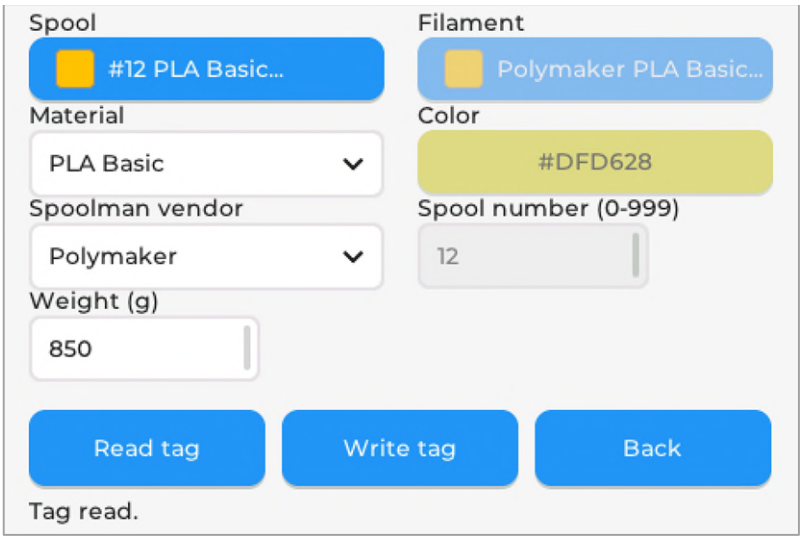


Figure 10: The editing screen with Spoolman

Field	Meaning
Spool	New spool or an existing spool from Spoolman. The list shows the newest spool at the top.
Filament	Only for a new spool: an existing filament from Spoolman that fills in material, colour and vendor.
Spoolman vendor	The vendor used for a new spool in Spoolman.



Figure 11: Choosing the spool

- If you choose an existing spool, the terminal fills in material, colour, vendor and weight from it and locks these fields. When writing, its Spoolman number is written to the tag.
- With New spool the terminal looks for a matching filament (vendor, material and colour) in Spoolman when writing and creates a new spool for it. Its number is then on the tag.
- If it finds no matching filament, or Spoolman cannot be reached, the tag is written without a link to Spoolman and the status line says so.
- When reading a tag, the terminal selects the matching spool straight away if Spoolman knows it.
- The Filament list shows the vendor, material and name of the filaments from Spoolman. The first time it is opened it takes a moment; a spinning circle is shown meanwhile.

- The A-Z button to the left of the list opens a keyboard of digits and letters. When you tap a character, the list shows only the filaments whose vendor starts with it; # stands for all other characters and All removes the filter. Characters that no vendor starts with are grayed out. The chosen character is kept the next time the list is opened.
- The top entry of the list is only a hint and is grayed out as long as no filament is chosen. Once you have chosen one, it reads (no template); tap it to drop the choice and free material, color and vendor again.



Figure 12: The list of filaments

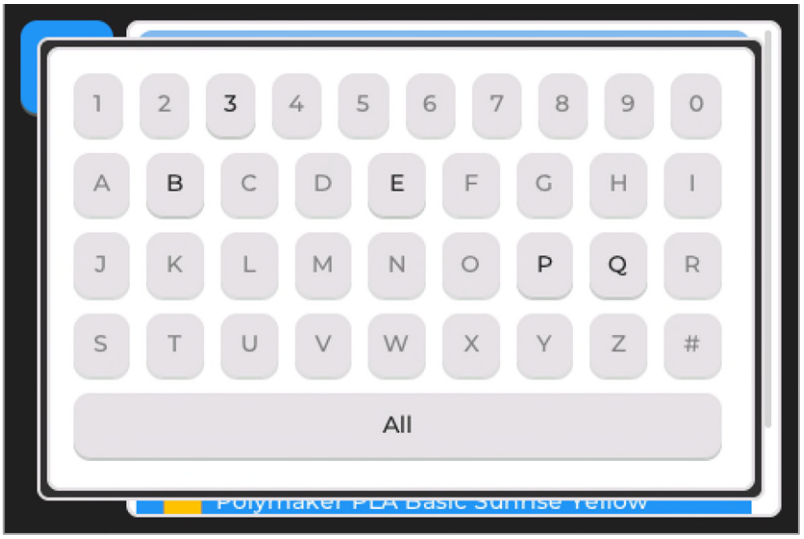


Figure 13: The keyboard for the filter



Figure 14: The filtered list (letter P)

6 The settings

The gear on the main screen opens the settings. They are divided into four tabs. Back returns to the main screen; changes are saved at once.

6.1 General

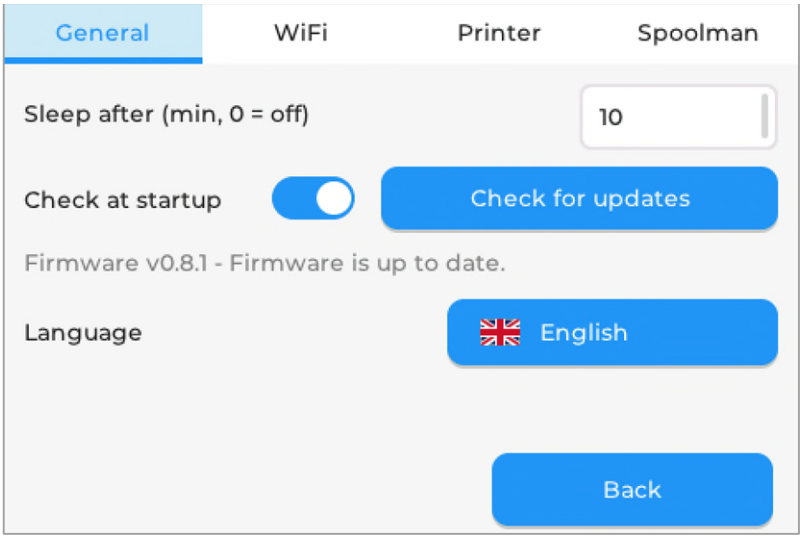


Figure 15: The settings, General tab

Setting	Meaning
Sleep after (min, 0 = off)	After this time without a touch the display goes dark. The next touch wakes it up; it does not press a button. A tag put on the reader also wakes the display. 0 switches sleeping off.
Check at startup	At startup the terminal looks for a new firmware (see Firmware update).
Check for updates	Looks for a new firmware right away. Below it is the version of the running firmware.
Language	Chooses the language of the user interface (see section Language).

6.2 WiFi

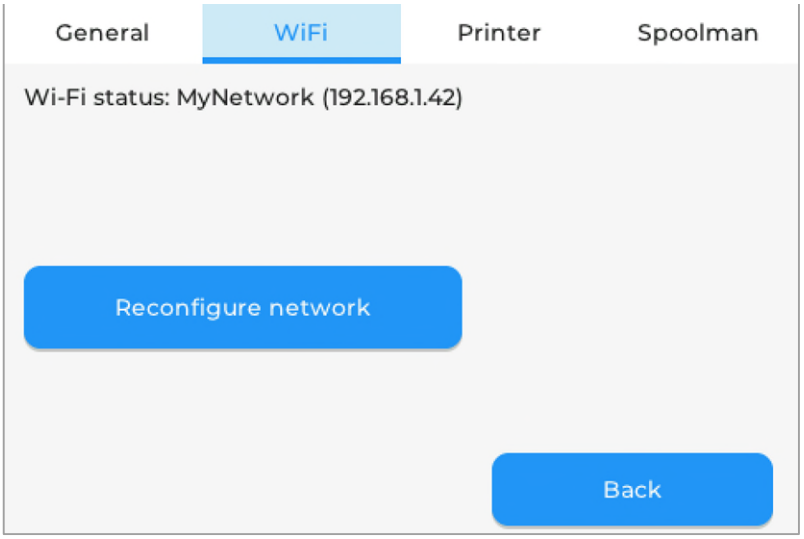


Figure 16: The settings, WiFi tab

The tab shows which Wi-Fi network the terminal is connected to, and its address. Reconfigure network deletes the saved Wi-Fi credentials and restarts the terminal. Then it opens the RFIDwisp-Setup Wi-Fi again as on the first start. The printers and the Spoolman settings are kept.

6.3 Printer

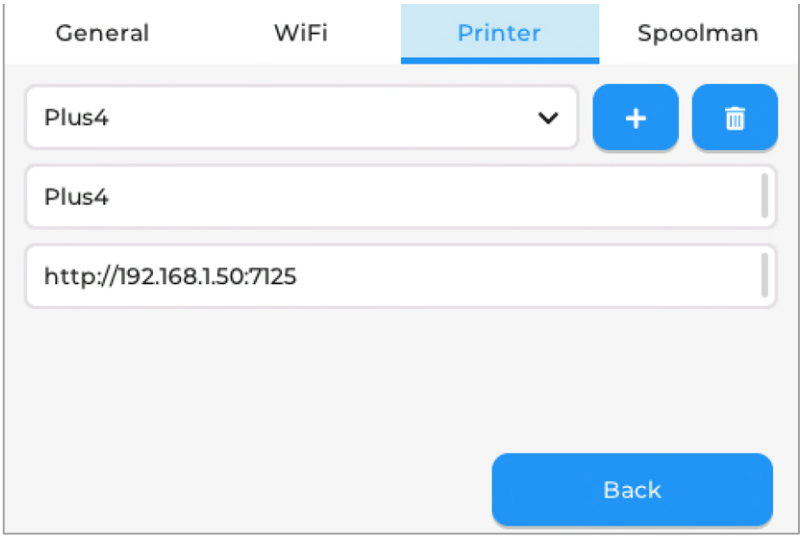


Figure 17: The settings, Printer tab

Here you manage the printers whose boxes appear on the main screen. Up to eight printers are possible.

- The list chooses the printer you edit. The plus adds a new printer, the waste bin deletes the chosen one.
- Name: the name under which the printer appears in the selection.

- Moonraker address: the address of the printer, for example `http://192.168.1.50:7125`. If `http://` is missing, the terminal adds it. An invalid address or a missing name is reported in red and not saved.
- A printer is only saved once name and address are filled in.

Tapping a text field brings up a keyboard at the bottom. The check mark confirms the entry.

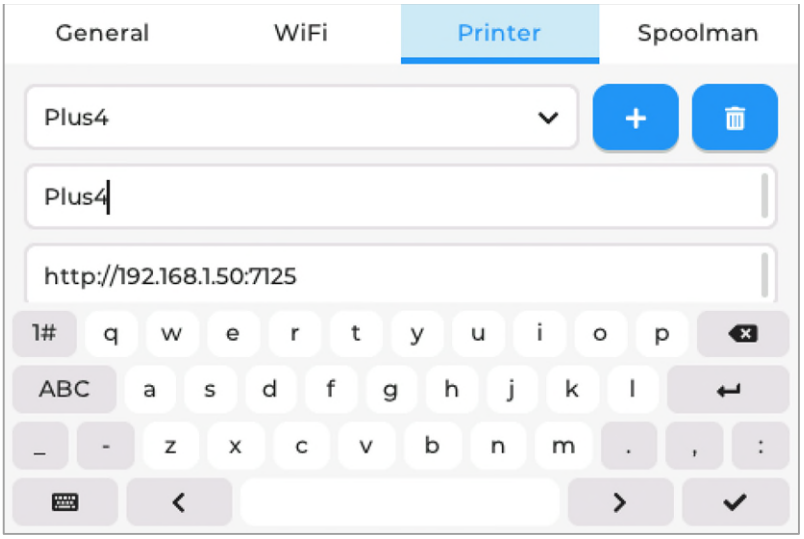


Figure 18: The keyboard while typing

6.4 Spoolman

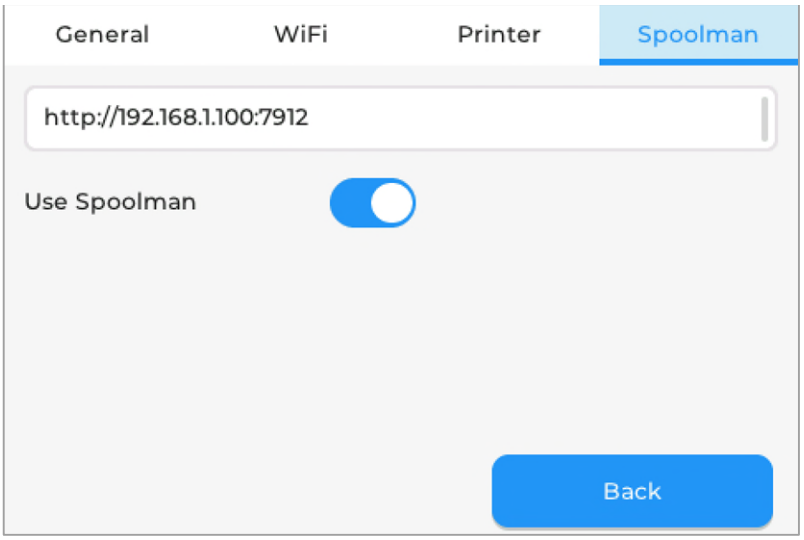


Figure 19: The settings, Spoolman tab

Enter the address of the Spoolman server, for example `http://192.168.1.100:7912`. Use Spoolman switches Spoolman on or off. When off, the terminal does not contact Spoolman at all; the editing screen then has the simple fields only.

7 Language

The user interface is available in German and English. In the General tab the Language button shows the current language with its flag. Tapping it opens the list of languages. After the choice the terminal restarts and then shows everything in the new language. The choice is kept in the device.

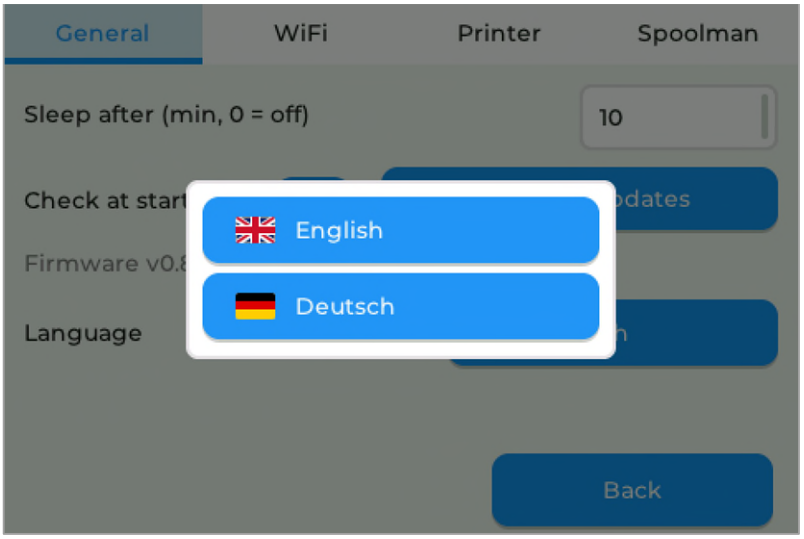


Figure 20: The list of languages

8 Firmware update

The terminal can update itself. It asks GitHub for the latest version (`https://github.com/ThorSc/RFIDwisp-Terminal/releases/latest`) and installs it if it is newer than the running one.

- If Check at startup is on and Wi-Fi is connected, the terminal looks at startup and asks whether to install the new version. Install starts the update, Later puts it off.
- With Check for updates in the settings you look right away. If a newer version was found, the same button is called Install with the version number.

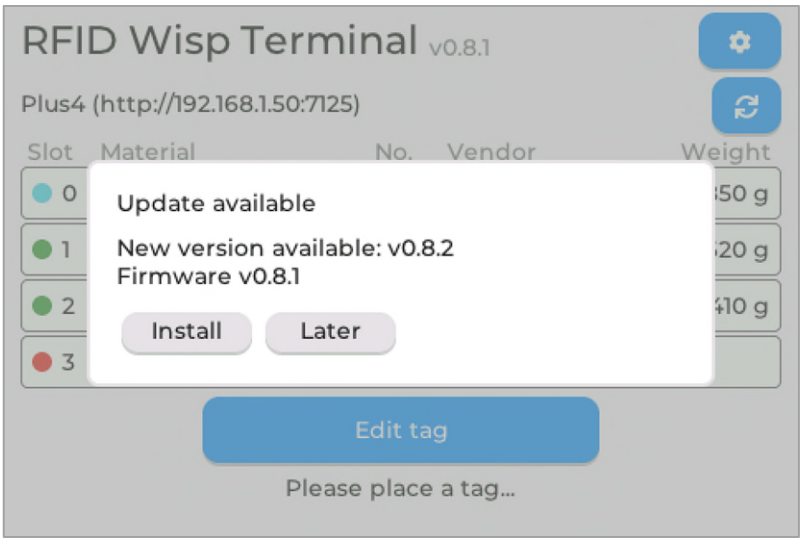


Figure 21: The question about installing

The terminal downloads the file securely over TLS, checks it against the checksum (SHA-256) published by GitHub and only then activates it. During the installation a screen shows the progress. Do not switch the terminal off meanwhile. Afterwards it restarts. An interrupted or faulty download leaves the running firmware unchanged.

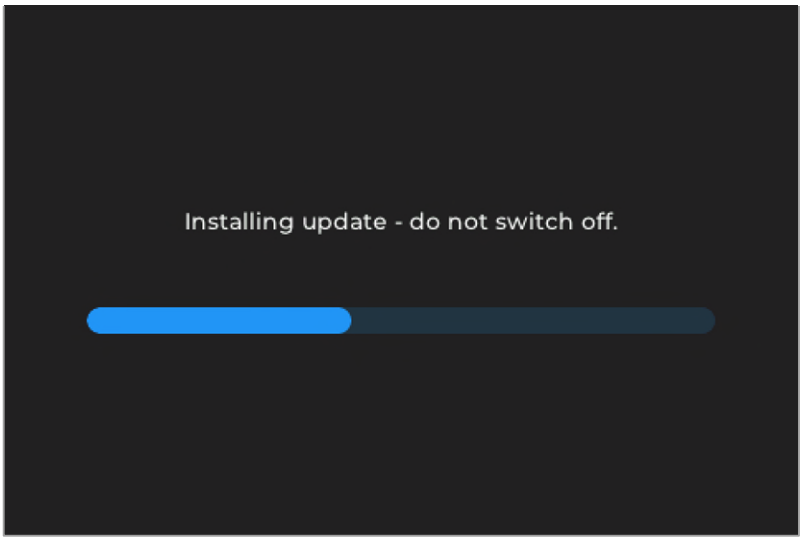


Figure 22: The installation is running

> **Note:** If you build the firmware yourself, you can also load it onto the terminal over the network (OTA without a cable); it announces itself as RFIDwisp-mobile in the network. The project's README has the details.

9 Troubleshooting

The status line at the bottom says what did not work. The most frequent messages:

Message	Cause and remedy
No tag present.	No tag was recognised within three seconds when reading or writing. Put the tag in the middle of the RFID module and try again.
Not a MIFARE Classic 1K tag.	The tag is of another kind. Use the tags of the QIDI spools or MIFARE Classic 1K tags.
Authentication failed (sector 1).	The tag cannot be opened with any of the known default keys. It is locked or belongs to another application.
Read failed. / Write failed or not confirmed.	The tag was not steady enough or too far away. Hold it still on the module during the operation.
Tag does not hold valid RFIDwisp spool data.	The tag is written, but not in the format of the spool tags. You can overwrite it.
RC522 not found.	The terminal cannot reach the RFID module. Check the wiring (see the README).
not connected	The printer does not answer. Check that it is switched on, the address is right and the terminal is on the Wi-Fi network. The reason is shown below the message, where the table is otherwise.
not supported	The printer has no QIDI box data. QIDI Plus4 and Max4 are supported.

Message	Cause and remedy
Could not load Spoolman data.	Spoolman cannot be reached. Check the address in the Spoolman tab and the server. The terminal then carries on with the simple fields.